

Part 1:**1. Why can the 2 routers see each other via pinging each other's IP address on the connected interface?**

Because the routers share the same direct connect network. This means they can communicate without special routing because the packets go back and forth

2. Why can one router NOT ping the other router's loopback interface?

Because the loop back addresses are different and the loop back addresses only go to the router it came from, hence loop back.

3. What is the purpose of configuring a loopback interface on a router?

Since the connection won't be lost if wires are unplugged, loopback interfaces on routers are used for identification, management access, and as a reference point for certain protocols

4. Screen print showing each router pinging the other's interface

```
alpine-host-1
/ # vysh
sh: vysh: not found
/ # ping 10.1.2.1
PING 10.1.2.1 (10.1.2.1): 56 data bytes
64 bytes from 10.1.2.1: seq=0 ttl=64 time=0.211 ms
64 bytes from 10.1.2.1: seq=1 ttl=64 time=0.096 ms
64 bytes from 10.1.2.1: seq=2 ttl=64 time=0.202 ms
^C
--- 10.1.2.1 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.096/0.169/0.211 ms
/ # ping 10.1.1.1
PING 10.1.1.1 (10.1.1.1): 56 data bytes
64 bytes from 10.1.1.1: seq=0 ttl=64 time=0.309 ms
64 bytes from 10.1.1.1: seq=1 ttl=64 time=0.120 ms
^C
--- 10.1.1.1 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.120/0.214/0.309 ms
/ # ping 1.1.1.1
PING 1.1.1.1 (1.1.1.1): 56 data bytes
64 bytes from 1.1.1.1: seq=0 ttl=64 time=0.324 ms
64 bytes from 1.1.1.1: seq=1 ttl=64 time=0.343 ms
64 bytes from 1.1.1.1: seq=2 ttl=64 time=0.350 ms
^C
--- 1.1.1.1 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.324/0.339/0.350 ms
/ # ping 10.1.1.2
PING 10.1.1.2 (10.1.1.2): 56 data bytes
64 bytes from 10.1.1.2: seq=0 ttl=63 time=0.809 ms
64 bytes from 10.1.1.2: seq=1 ttl=63 time=0.257 ms
64 bytes from 10.1.1.2: seq=2 ttl=63 time=0.247 ms
^C
```

```
alpine-host-1
64 bytes from 1.1.1.1: seq=2 ttl=64 time=0.350 ms
^C
--- 1.1.1.1 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.324/0.339/0.350 ms
/ # ping 10.1.1.2
PING 10.1.1.2 (10.1.1.2): 56 data bytes
64 bytes from 10.1.1.2: seq=0 ttl=63 time=0.809 ms
64 bytes from 10.1.1.2: seq=1 ttl=63 time=0.257 ms
64 bytes from 10.1.1.2: seq=2 ttl=63 time=0.247 ms
^C
--- 10.1.1.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.247/0.437/0.809 ms
/ # ping 10.1.3.1
PING 10.1.3.1 (10.1.3.1): 56 data bytes
64 bytes from 10.1.3.1: seq=0 ttl=63 time=0.445 ms
64 bytes from 10.1.3.1: seq=1 ttl=63 time=0.212 ms
^C
--- 10.1.3.1 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.212/0.328/0.445 ms
/ # 10.1.3.10
sh: 10.1.3.10: not found
/ # ping 10.1.3.10
PING 10.1.3.10 (10.1.3.10): 56 data bytes
64 bytes from 10.1.3.10: seq=0 ttl=62 time=0.460 ms
64 bytes from 10.1.3.10: seq=1 ttl=62 time=0.215 ms
64 bytes from 10.1.3.10: seq=2 ttl=62 time=0.298 ms
^C
--- 10.1.3.10 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.215/0.324/0.460 ms
/ #
```

```
alpine-host-2
Trying 127.0.0.1...
Connected to localhost.
Escape character is '^['.

BusyBox v1.30.1 (Ubuntu 1:1.30.1-7ubuntu3) built-in shell (ash)
Enter 'help' for a list of built-in commands.

/ # ping 10.1.3.1
PING 10.1.3.1 (10.1.3.1): 56 data bytes
64 bytes from 10.1.3.1: seq=0 ttl=64 time=0.240 ms
64 bytes from 10.1.3.1: seq=1 ttl=64 time=0.110 ms
^Z[1]+  Stopped                  ping 10.1.3.1
/ # ^C
/ # ping 10.1.1.2
PING 10.1.1.2 (10.1.1.2): 56 data bytes
64 bytes from 10.1.1.2: seq=0 ttl=64 time=0.113 ms
64 bytes from 10.1.1.2: seq=1 ttl=64 time=0.195 ms
^C
--- 10.1.1.2 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.113/0.154/0.195 ms
/ # ping 2.2.2.2
PING 2.2.2.2 (2.2.2.2): 56 data bytes
64 bytes from 2.2.2.2: seq=0 ttl=64 time=0.209 ms
64 bytes from 2.2.2.2: seq=1 ttl=64 time=0.197 ms
^C
--- 2.2.2.2 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.197/0.203/0.209 ms
/ # ping 10.1.1.1
PING 10.1.1.1 (10.1.1.1): 56 data bytes
64 bytes from 10.1.1.1: seq=0 ttl=63 time=0.835 ms
64 bytes from 10.1.1.1: seq=1 ttl=63 time=0.353 ms
^C
```

```
alpine-host-2
/ # ping 2.2.2.2
PING 2.2.2.2 (2.2.2.2): 56 data bytes
64 bytes from 2.2.2.2: seq=0 ttl=64 time=0.209 ms
64 bytes from 2.2.2.2: seq=1 ttl=64 time=0.197 ms
^C
--- 2.2.2.2 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.197/0.203/0.209 ms
/ # ping 10.1.1.1
PING 10.1.1.1 (10.1.1.1): 56 data bytes
64 bytes from 10.1.1.1: seq=0 ttl=63 time=0.835 ms
64 bytes from 10.1.1.1: seq=1 ttl=63 time=0.353 ms
^C
--- 10.1.1.1 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.353/0.594/0.835 ms
/ # ping 10.1.2.1
PING 10.1.2.1 (10.1.2.1): 56 data bytes
64 bytes from 10.1.2.1: seq=0 ttl=63 time=0.551 ms
64 bytes from 10.1.2.1: seq=1 ttl=63 time=0.183 ms
^C
--- 10.1.2.1 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.183/0.367/0.551 ms
/ # ping 10.1.2.10
PING 10.1.2.10 (10.1.2.10): 56 data bytes
64 bytes from 10.1.2.10: seq=0 ttl=62 time=0.668 ms
64 bytes from 10.1.2.10: seq=1 ttl=62 time=0.350 ms
64 bytes from 10.1.2.10: seq=2 ttl=62 time=0.301 ms
64 bytes from 10.1.2.10: seq=3 ttl=62 time=0.322 ms
^C
--- 10.1.2.10 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 0.301/0.410/0.668 ms
/ #
```

5. Why did we have to add static routes, and what do they do?

Static routes are additional routes for the router as the router only knows the networks directly connected to itself. The static routes used tell the router how to reach a network that is connected to a network that the router is directly connected to.

Part 2:

1. A lot of magic happened with the commands `router ospf`, and `neighbor 0.0.0.0/0 area 0`.

What exactly do you think these commands do?

router ospf enables the OSPF routing process on the router, which tells the router to start running the OSPF protocol, which is a link state routing algorithm that lets routers dynamically discover and share routing information with each other.

network 0.0.0.0/0 area 0 tells OSPF to display all the router's interfaces into OSPF Area 0. The 0.0.0.0/0 is an address(?) that matches every IP address, so every interface is OSPF-enabled.

When both routers do this, they send Hello packets to find each other, form neighbor adjacency, exchange link state databases, and then independently calculate the best paths.

2. There are 3 basic subnetworks in this lab. 10.1.1.0/30, 10.1.2.0/24, 10.1.3.0/24. Why are we using a /30 between the routers?, Why cant host-1 and host-2 be on the same Network?

A /30 subnet gives 2 usable host addresses which is what is needed for a point to point link between two routers. Using a larger subnet like /24 would waste addresses on a link that only ever has two endpoints. The hosts can't be on the same network because they are on opposite sides of the two routers, physically separated by the router to router link. Routing only occurs between different networks. Putting them on separate subnets makes their traffic be routed through R1 and R2, which is what OSPF is configured to handle.

3. Provide your configuration for each of the routers. You can get this by typing “do show running-config”

```
R1# config t
R1(config)# do show running-config
Building configuration...

Current configuration:
!
frr version 8.4.1_git
frr defaults traditional
hostname R1
no ipv6 forwarding
service integrated-vtysh-config
!
interface eth0
 ip address 10.1.1.1/30
exit
!
interface eth1
 ip address 10.1.2.1/24
exit
!
interface lo
 ip address 1.1.1.1/32
exit
!
router ospf
 network 0.0.0.0/0 area 0
exit
!
end

R2# config t
R2(config)# do show running-config
Building configuration...

Current configuration:
!
frr version 8.4.1_git
frr defaults traditional
hostname R2
no ipv6 forwarding
service integrated-vtysh-config
!
interface eth0
 ip address 10.1.1.2/30
exit
!
interface eth1
 ip address 10.1.3.0/24
 ip address 10.1.3.1/24
exit
!
interface lo
 ip address 1.1.1.2/32
exit
!
router ospf
 network 0.0.0.0/0 area 0
exit
!
end
R2(config)#
```